

1

Get your station ready

How do three different liquids change a hydrogel?

Compare vinegar, DI water, and baking-soda solution. Run all three conditions in each of three trials.



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YOUR GROUP NEEDS

3 test containers + 1 BASIC mixing beaker

1 graduated cylinder; 3 labeled spoons; pH strips

Goggles; timer; marker; paper towels; data sheet

9 portions of 0.50 g hydrogel; 1 packet of 1.5 g baking soda

DI water and instructor-supplied vinegar

Check the containers before you pour.

Each sample receives $40 + 50 = 90$ mL of liquid. Use containers with comfortable stirring space; ask your instructor to check the Petri dishes first. A 150 mL or larger cup/beaker is a practical alternative.

Wear goggles. Keep powder away from your face. Do not taste any material. Tell your instructor about spills.

2

Prepare and label

Make BASIC once for all three trials.

1.5 g

baking soda

+

150 mL

DI water

1. Empty the baking-soda packet into the mixing beaker.
2. Measure 150 mL DI water with the cylinder and add it.
3. Stir until dissolved. Label the beaker BASIC.

Reserve this solution: use 50 mL in each trial.

ACIDIC

NEUTRAL

BASIC

Label your three test containers and spoons to match. Keep the mixing beaker separate from the BASIC test container.

Check pH before adding solutions to the gel.

Use a fresh pH strip for each test liquid, follow the strip instructions, and record the reading. Read the actual pH; do not assume a number from the label.

DI = deionized water. Use the supplied water and vinegar. Rinse the measuring cylinder with DI water and drain between different liquids.

3

Make three matching gels

Use this recipe in EACH labeled container.

ACIDIC**0.50 g**

hydrogel powder

+ 40 mL

DI water

NEUTRAL**0.50 g**

hydrogel powder

+ 40 mL

DI water

BASIC**0.50 g**

hydrogel powder

+ 40 mL

DI water

1. Add one 0.50 g portion of hydrogel powder to each container. Use pre-portioned powder or the instructor's balance.
2. Measure and add 40 mL DI water to each container. Read the bottom of the water curve at eye level.
3. Stir gently as each gel forms. Use the same preparation method for all three. Note how the gels look before testing.

Pause before the test liquids.

Ask your instructor to confirm the gels are ready. If a sample is still mostly dry or looks very different, ask before changing its recipe.

4

Add, stir, then wait

Match each liquid to its label.

Container	Add to the prepared gel
ACIDIC	50 mL supplied vinegar
NEUTRAL	50 mL DI water
BASIC	50 mL prepared BASIC solution

Keep spoons with their matching containers. Rinse and drain the cylinder between different liquids.

10 stirs

same gentle motion

5 minutes

after that sample is stirred

After adding the test liquid, stir that sample 10 times. Start its 5-minute wait as soon as its stirring is complete.

Give every sample the same waiting time.

Use separate timers, write down each start time, or test one condition at a time. Score each sample after its own 5-minute wait.

5**Score, record, repeat**

Use the same 1-5 scale for every sample.

5**Fully solid / no change****4****Slightly smaller / softened****3****Partly liquid****2****Mostly liquid****1****Fully liquid**

Compare with the gel before the test liquid was added. Agree on the closest score, then record a brief observation. Similar scores are valid results.

One trial = all three conditions.

Trial 1: Acidic + Neutral + Basic

Trial 2: three fresh gel samples

Trial 3: three fresh gel samples

Between trials: record first, then scrape the gel into the instructor-designated solid-waste bin. Wipe out the containers; rinse/drain as directed. Reuse the clean containers and repeat Steps 3-5.

6 Your results and cleanup

Group: _____ Date: _____

Condition	pH of test liquid
Acidic	
Neutral	
Basic	

Condition	Trial 1	Trial 2	Trial 3	Mean	SD*
Acidic					
Neutral					
Basic					

Mean = (Trial 1 + Trial 2 + Trial 3) / 3

Example scores 2, 3, 4: mean = 3.0. *If your instructor requests sample standard deviation: subtract the mean from each score, square the three differences, add them, divide by 2, then take the square root. For this example, SD = 1.0.

What changed? Which results were most consistent?

Finish: put gel in the assigned solid-waste bin, not the sink. Follow instructor directions for leftover liquids and rinsing. Clean the bench, return equipment, and wash your hands.